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Distribution of plastic strain in cutting of Ti-6Al-4V titanium alloy using experimental and modelling approaches

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Abstract

Metal cutting is considered a process of severe plastic deformation (SPD) and often compared to other SPD processes used to develop submicron and nanostructured metallurgical structure in metallic materials. To validate this assumption, a proper determination of the distribution of plastic strain in the machined surface and subsurface is required. This research work deals with the determination of the distribution of the plastic strain in the machined surface and subsurface in orthogonal cutting of Ti-6Al-4V titanium alloy using both experimental and modeling approaches. Plastic strain distribution in the machined subsurface is determined experimentally using in-situ Digital Image Correlation (DIC), while numerical simulation using FEM is used to determine plastic strain distribution up to the machined surface. The experimental plastic strain distribution obtained by DIC and by X-ray diffraction peak broadening is used to validate the predicted plastic strain distribution obtained by FEM simulation. The results show a strong gradient of plastic strain in a thin layer less than 50 μm , with high plastic strain values at the machined surface exceeding 50%.

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1. Introduction

Deformation processes are a large group of manufacturing processes involving the transformation of an initial relatively simple shape (billet and sheet blank) into a more complex shape by plastic deformation using specific tools (cutting tools, molds and dies). Metal forming (forging, extrusion and stamping) are the most known deforming processes, but metal cutting can also be considered as a deformation process, since it also involves plastic deformation of the work material using the cutting tools to produce a part with a given shape. The principal difference between metal cutting and all other deformation processes is the physical separation of the layer being removed in the form of chips from the rest of the workpiece. The process of physical separation of a solid body into two or more parts is known as fracture, and thus, metal cutting must be treated as the purposeful fracture of the layer

being removed. This fracture occurs due to the continuous change of the state-of-stress in the first deformation zone causing a cyclic nature of the process [1]. The plastic deformation can be beneficial or detrimental for the process efficiency and part quality, depending on the process goal. For metal forming processes the plastic deformation should be sufficiently high to obtain a suitable part shape without fracture. The opposite is true for metal cutting. As mentioned by Astakhov and Shvets [2], the main goal in metal cutting should be the minimization of the external energy applied to the cutting system by generating suitable state-of-stresses that reduces the plastic deformation, thus the energy required for this material separation. Therefore, the energy spent on plastic deformation in metal cutting must be considered as wasted.

Metal cutting processing is accomplished with low or significant degree of plastic deformation depending on their properties, but also on the processing conditions. For a given

material, the processing or cutting conditions (cutting regime parameters, tool geometry and tool material, lubrication/cooling, etc) will affect the state-of-stress [3], thus the strain at fracture [4], and consequently the amount of plastic deformation (thus energy) required for the material separation. Abushawashi et al. [3] investigated the influence of the cutting conditions in the stress triaxiality in the first deformation zone (chip formation) in orthogonal cutting simulation of AISI 1045 steel. For instance, they found that stress triaxiality increases significantly when the rake angle is reduced, thus increasing the strain at fracture of the work material. This simulation result is confirmed by the experimental values of the chip compression ratio (CCR), the ratio of the chip thickness to the uncut chip thickness, where this parameter increases as the rake angle decreases. As pointed out by Astakhov and Shvets [2], CCR (when properly measured) directly reflects the final plastic deformation that takes place in the cutting process. The main goal of metal cutting studies should be the optimization of the cutting conditions to minimize the amount of plastic strain, thus the amount of energy required to separate the material. The particular case of Ti-6Al-4V used to develop submicron and nanostructured metallurgical structure when undergoing severe plastic deformation induced by machining [5-6]. To validate this assumption, in this study, we determine the distribution of the plastic strain in the machined surface and subsurface in orthogonal cutting of Ti-6Al-4V titanium alloy using both experimental and modeling approaches to quantify the strong gradient of plastic strain in a thin layer induced by the machining process.

2. Experimental setup and cutting conditions

Ti6Al4V titanium alloy used in this study is composed of two-phase structure ($\alpha + \beta$). The good balance between α and β phases makes it achieve high strength level without losing ductility [7]. The average microhardness, hardness and grain size of this alloy are 330 ± 18 HV, 303 ± 8 HBW and 15.5 ± 1.8 μm , respectively.

Orthogonal cutting tests were performed over rectangular specimens (40mm(L) x 20mm(H) x 4mm(W)) in Ti6Al4V alloy using uncoated cemented carbide (WC-Co) cutting tools. Tool geometry in the tool-in-hand system according to the ISO standard 3002:1982 was checked carefully using Alicona Infinite Focus SL microscope. Each insert was placed in the holder to measure the tool rake, γ_n , the tool clearance, α_n , and the cutting-edge radius, r_n .

The results of these measurements are shown in Table 1 together with the corresponding values of the cutting regime parameters (cutting speed V_c , uncut chip thickness h , and width of cut w). All the tests were conducted under dry cutting conditions.

Table 1. Cutting conditions used in orthogonal cutting test.

Cutting parameters	Value
Cutting speed, V_c (m/min)	20; 60
Uncut chip thickness, h (mm)	0.15; 0.2
Tool cutting edge radius, r_n (μm)	16 ± 1.1 ; 30 ± 1.9
Tool rake angle, γ_n ($^\circ$)	5 ± 0.13 ; -6 ± 0.25
Tool clearance angle, α_n ($^\circ$)	6 ± 0.23
Width of cut, w (mm)	4
Cooling condition	Dry cutting

Orthogonal cutting tests in planning configuration were performed in a 3 axis CNC milling machine from DMG model

DMC85V, as can be seen in Fig. 1a. The tool was clamped in a square supporter which is fixed on the machine head.

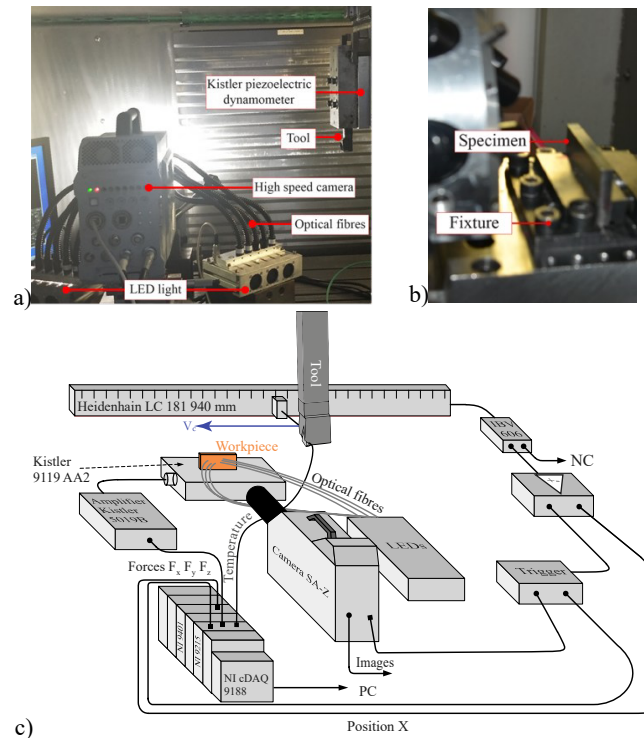


Fig. 1: Experimental setup to conduct the orthogonal cutting tests.

This milling machine is equipped with a linear motor to generate a cutting speed up to 120 m/min and an acceleration up to 20 m/s^2 . A gradual acceleration has been performed by a speed loop-back command to reach the specified cutting speed in some millimeters before interaction between tool and workpiece. Each test was performed using a new cutting edge to eliminate the tool wear effect. Fig. 1c shows the schematic representation of the acquisition system to measure the forces and the equivalent strain distribution during cutting. In this paper, only strain results are exploited. The cutting zone of the specimen was observed by a high-speed CCD camera from Photron, model Fastcam SA-Z. Framerate is set at 20,000 fps, ensuring an image acquisition each 0.05 ms (approximately 50 μm travelled by the cutting tool for a cutting of 60 m/min) with an exposure time of $1/80,000$ sec. To get a high magnification of the cutting zone and machined surface, $\times 10$ magnification Mitutoyo objective set together with extension lens tubes are assembled to the camera. A $1.64 \times 1.64 \text{ mm}^2$ observation area is obtained from an image of 1024×1024 square pixels resolution with 8 bits dynamic range (256 gray levels), and a pixel calibration of $1.6 \mu\text{m}/\text{pixel}$. These conditions ensured the acquisition of enough images for the Digital Image Correlation (DIC) [8].

Additionally, to get a clear detection of the tool in the images taken by the camera, the insert has been grounded using a grinding machine to obtain a flat surface normal to the camera axis. Meanwhile, the lateral surface of specimen under observation by the camera was sandblasted to generate a suitable pattern for DIC analysis, using glass micro-spheres with diameters between 50 and $100 \mu\text{m}$. The surface quality was determined by two parameters: sandblasting time and pressure. To choose the appropriate parameters, five combinations of pressure and time are evaluated. The optimum

sandblasting parameters is the one that gives the widest grey level on the surface under observation by the camera, which corresponds to a pressure of 4 bar and a time of 9 sec. These parameters were used for sandblasting all the samples used in the machining tests.

DIC software CorreliQ4 was used to calculate the displacements and strains distributions in the cutting zone of the samples, based on the images acquired by the high-speed camera. This software uses a finite element type global approach based on Q4P1 shape functions as described by Hild and Roux [9]. This FE-based global approach can capture complex displacement fields and leads to high resolution. In addition, it is directly linked to numerical simulations without any interpolation of the measured displacements. The equivalent strain is calculated by the following formula [2]:

$$\epsilon_{eq} = \frac{\sqrt{2}}{3} \left[(\epsilon_x - \epsilon_y)^2 + (\epsilon_y - \epsilon_z)^2 + (\epsilon_z - \epsilon_x)^2 + 6(\epsilon_{xy}^2 + \epsilon_{yz}^2 + \epsilon_{zx}^2) \right]^{\frac{1}{2}} \quad (1)$$

Cutting force F_c and thrust force F_T were measured by a piezoelectric dynamometer from Kistler, model 9119AA2, together with a charge amplifier Kistler, model 5019B.

Finally, Full Width at Half Maximum (FWHM) [10] were measured at the machined surface of the samples using the X-ray diffraction technique in the longitudinal (x-direction in [10]) and transversal (y-direction) directions using Seifert XRD 3000 PTS equipment. Measurements were made with the diffraction of copper K- α radiation from the {213} crystallographic plane of the Ti6Al4V alloy. Diffractograms were recorded for 29 tilt angles ψ varying between -45° and $+45^\circ$. Measurements were performed in the center of the machined surface. The irradiation zone had a length of 4 mm and width 2 mm. The penetration depth of the X-Ray radiation was about 5 μm . To measure the FWHM beneath the surface, the layers were successively removed by electrolytic polishing method using Presi Poliselec C20 equipment, to avoid reintroducing additional residual stresses.

3. Orthogonal cutting model

3.1. General description

2D orthogonal cutting model of Ti-6Al-4V titanium alloy is developed and simulated using Lagrangian approach in ABAQUS/Explicit FEA software. This model and corresponding boundary conditions is shown in Fig.2. It is composed by the workpiece (1) and tool (2). The workpiece is divided into three zones: upper layer used to form the chip (1a), intermediate layer (1b), and bottom layer that corresponds to the machined workpiece (1c). The intermediate layer is applied to simulate the physical separation of the layer being removed (in the form of chips) from the rest of workpiece, using the element deletion technique. The workpiece is fixed in the bottom in both X and Y directions, while the tool is moving along the negative X-direction at a speed equal to the cutting speed (V_c) and fixed in the Y-direction. The bottom and right sides of the workpiece together with the right and upper sides of the tool are at room temperature of 20 °C. To simulate the dry cutting conditions, the other sides of the workpiece and tool

are convective surfaces with the air, with a coefficient of heat convection of 23 W/m²/°C [11]. An initial temperature of 20 °C is imposed to both the tool and the workpiece.

The quadrilateral continuum elements CPE4RT are used. Refined elements are used in the parts of chip and intermediate layer to better simulate the cutting process, where the mesh size is 17 μm x 10 μm . Meanwhile, the mesh along the depth of workpiece surface is also refined to calculate the distribution of residual stress, where the mesh size was defined as 17 μm x 7.5 μm .

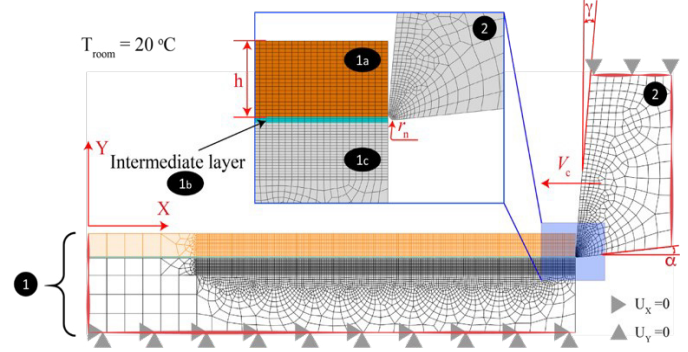


Fig. 2: 2D orthogonal cutting model.

A coupled thermo-mechanical analysis was performed. Temperature dependent thermal, elastic, and physical properties of Ti-6Al-4V alloy and tool are provided by the titanium alloy supplier and tool maker, completed by data taken from the Outeiro et al. 2015 [12]. Reference values of these properties at room temperature are given in Table 2.

Table 2. Thermal, elastic, and physical properties of the workpiece and tool at room temperature (20 °C).

Physical properties	Workpiece	Tool
Density, ρ (kg/m ³)	4420	13967
Elastic modulus, E (GPa)	114	627.5
Poisson's ration, ν	0.31	0.25
Specific heat, C_p (J/kg ^o C)	670	100
Thermal conductivity, λ (W/m ^o C)	6.6	9.7
Expansion coefficient, α_v ($\mu\text{m}/\text{m}^\circ\text{C}$)	9	5.174

3.2. Material constitutive and contact models

As far as the constitutive model of Ti-6Al-4V alloy is concerned, a model proposed by Cheng et al. [13] is used and implemented in Abaqus through VUMAT and UMAT subroutines. This model includes both plasticity and damage formulations. The plasticity model incorporates the strain hardening, strain rate and state of stress effects, and is given by equations 2, 3 and 4.

$$\bar{\sigma} = (A + m\epsilon_p^n) \left[B + C \ln \left(E + \frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right] \left[1 - c_\eta (\eta - \eta_0) \right] \left[c_\theta^s + (c_\theta^{ax} - c_\theta^s) \left(\gamma - \frac{\gamma^{a+1}}{a+1} \right) \right] \quad (2)$$

$$\gamma = 6.464 \left[\sec \left(\frac{\bar{\theta}\pi}{6} \right) - 1 \right] \quad (3)$$

$$c_{\theta}^{\text{ax}} = \begin{cases} c_{\theta}^t & \text{for } \bar{\theta} > 0 \\ c_{\theta}^c & \text{for } \bar{\theta} \leq 0 \end{cases} \quad (4)$$

In the previous equations: i) the coefficients A , m and n are used in the strain hardening term; ii) the coefficients B , C and E are used in the strain rate term; iii) the coefficient c_{η} is used in the stress triaxiality term; iv) the coefficients c_{θ}^t , c_{θ}^c , c_{θ}^s and a are used in the Lode angle term; v) η_0 is the reference stress triaxiality and $\dot{\epsilon}_0$ is the reference strain rate; vi) γ describes the difference between Tresca and von-Mises equivalent stresses on the deviatoric stress plane, as represented in Equation (5); and vii) the coefficients c_{θ}^t , c_{θ}^c , c_{θ}^s are interdependent and at least one of them equals 1. Considering the compression test of cylindrical specimen at quasi-static conditions as a reference, η_0 equals $-1/3$, $\dot{\epsilon}_0$ equals 0.05 s^{-1} and c_{θ}^c equals 1. The temperature effect on the work material behaviour is not considered in this plasticity model for the main reasons mentioned by Cheng et al. [13].

The damage model includes damage initiation and damage evolution, as shown from Eq. 5 to Eq. 7. Eq. 5 represents the damage initiation, while Eq. 6 and Eq. 7 represent the damage evolution.

$$\bar{\epsilon}_i^p = \left\{ \left[\frac{1}{2} (D_1 e^{-D_2 \eta} + D_3 e^{-D_3 \eta}) - D_3 e^{-D_4 \eta} \right] \bar{\theta}^2 + \frac{1}{2} (D_1 e^{-D_2 \eta} - D_3 e^{-D_3 \eta}) \bar{\theta} + D_3 e^{-D_4 \eta} \right\} \times \left[1 + D_7 \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right) \right] \quad (5)$$

$$\bar{\sigma} = (1 - D) \bar{\sigma}_0, \quad D = \frac{1 - \exp(\lambda \epsilon^*)}{1 - \exp(\lambda)}, \quad \epsilon^* = \frac{\bar{\epsilon}^p - \bar{\epsilon}_i^p}{\bar{\epsilon}_f^p - \bar{\epsilon}_i^p} \quad (6)$$

$$G_f = \int_{\bar{\epsilon}_i^p}^{\bar{\epsilon}_f^p} l \bar{\sigma} d\bar{\epsilon}^p = \int_0^{\bar{u}_f} \bar{\sigma} d\bar{u} \quad (7)$$

In these equations, D_1 - D_7 , n , and G_f are the material coefficients, λ controls the material degradation rate and G_f is the material fracture energy density; $\bar{\epsilon}_i^p$ is the plastic strain at damage initiation; $\bar{\epsilon}_f^p$ is the plastic strain when all the stiffness and the fracture energy of the material have been lost and dissipated, respectively; $\bar{\sigma}$ is the hypothetical undamaged stress evaluated by Equation (2); l is the characteristic length of the finite element; $\bar{\mu} = 0$ relates to the equivalent plastic displacement before damage initiates; and $\bar{\mu}_f$ is the equivalent plastic displacement at complete fracture. The coefficients of both plasticity and damage initiation models are given in the publication of Cheng et al. [13] respectively. In the case of damage evolution, G_f and λ equal to 18.5 KJ/m^2 and 9.4 , respectively [13].

Zorev's model [14] is adopted in this work to describe the contact stresses at tool-workpiece interface. Mathematically, these contacts stresses can be represented by Eq. 8:

$$\tau = \begin{cases} \mu \cdot \sigma & \text{if } \mu \cdot \sigma < \tau_{lim} \\ \tau_{lim} & \text{if } \mu \cdot \sigma \geq \tau_{lim} \end{cases} \quad (8)$$

where σ and τ are the normal and shear stresses, and μ is the friction coefficient at the interface. Therefore, for sliding contact conditions at the interface ($\mu \sigma < \tau_{lim}$), the shear stress will be calculated using the Coulomb friction model ($\tau = \mu \sigma$), while for sticking contact conditions at the interface ($\mu \sigma \geq \tau_{lim}$), the sliding does not occur at the tool-chip interface but in the adjacent layers of the chip, being the shear stress equal to the shear stress limit, τ_{lim} . This shear stress limit can be determined by $\sigma_y / \sqrt{3}$, where σ_y is the yield stress of the workpiece and chip materials, which is calculated during the simulation of the machining. The value of the friction coefficients between the Ti6Al4V and the uncoated tungsten carbide are determined by tribological tests under identical contact conditions as those applied during machining, concerning to sliding speed, pressure, temperature, lubrication, etc. They are as a function of sliding velocities, and given by Eq. 10, where $a = 0.3$ and $b = -0.0303$.

$$\mu = a + b \ln V_s \quad (9)$$

4. Comparison between measured and simulated results

The plastic strains were determined using the DIC technique. For each machining test a sequence of images of the lateral face of the workpiece under observation by the high-speed camera are taken. These images are taken before cutting (before the cutting tool be in contact with the workpiece), during cutting (when the tool is in contact with the workpiece creating an elastic-plastic deformation zone ahead of the cutting edge as the tool moves in the cutting direction), and after cutting (when the tool is no more in contact with the workpiece). Fig. 3 shows an example of three image taken before, during and after cutting, respectively. As described by Outeiro et al. [15], only three pairs of images are required to determine the plastic strain in the machined subsurface for each cutting condition.

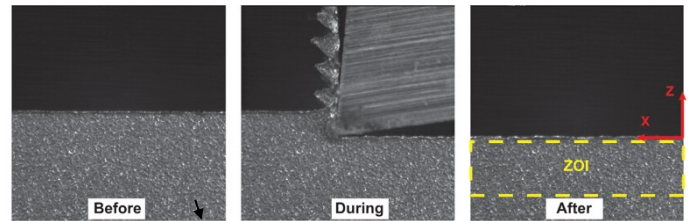


Fig. 3. Images taken in the orthogonal cutting tests: (a) before; (b) during; (c) after the contact between the tool and the workpiece.

A thin layer near the machined surface is not included in the ZOI due to the difficulties in determining the material displacement in this region. The equivalent plastic strains were calculated beneath the machined surface for several cutting conditions using Eq. 1. Fig. 4 shows an example of the distribution of the equivalent plastic strain in the zone of interest ZOI of the machined subsurface of the workpiece. The residual equivalent plastic strains in function of the depth from the machined surface (z -direction) for different cutting conditions are presented in figure 5. Each curve corresponds to an average of 10 curves taken along the cutting direction. This figure shows that the tool with positive rake angle generated lower maximum plastic strain near the surface than the negative one. This is according to what has been observed in metal

cutting, in particular the decrease of the CCR when the tool rake angle increases [2]. Moreover, this figure also shows that the thickness of the plastically deformed layer is varying from 80 to 160 μm .

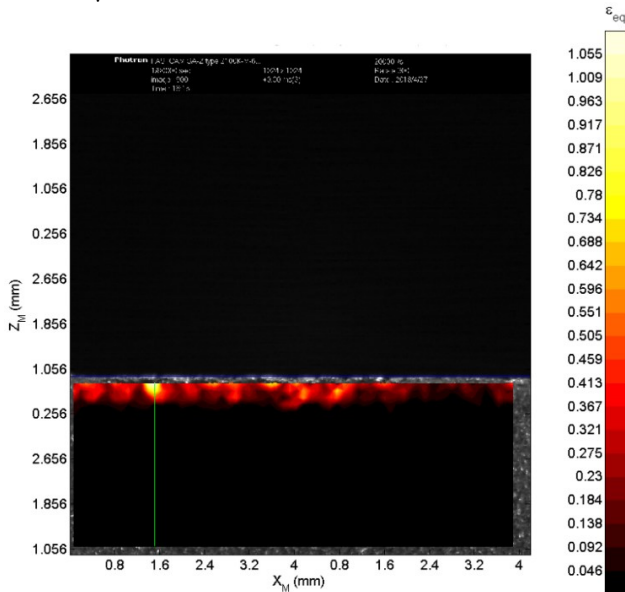


Fig. 4. Distribution of residual equivalent plastic strain ($\gamma = -6^\circ$, $V_c = 20$ m/min, $h = 0.2$ mm, $m = 30$ μm).

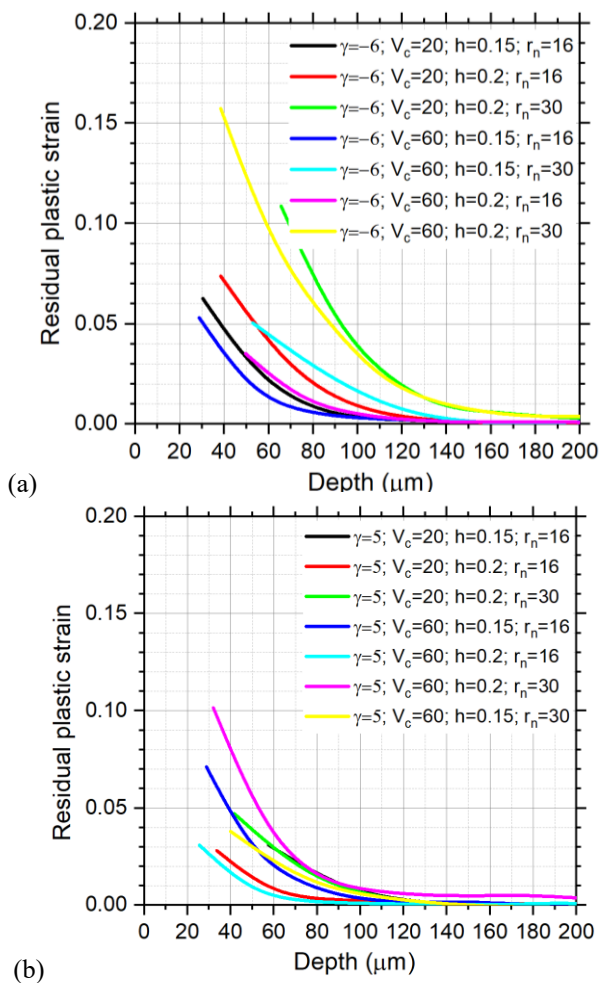


Fig. 5. Experimental profiles of residual equivalent plastic strain obtained by DIC for different cutting conditions, using (a) negative (-6°) and (b) positive (5°) tool rake angle.

Another method that can be used to determine the thickness of the plastically deformed layer induced by machining is the FWHM of the X-ray diffraction peak. Fig. 6 shows the evolution of the FWHM in function of the depth beneath the surface. FWHM decrease as the depth increases, stabilizing around 1.35° . The depth at which the FWHM stabilizes corresponds to the thickness of the plastically deformed layer induced by machining.

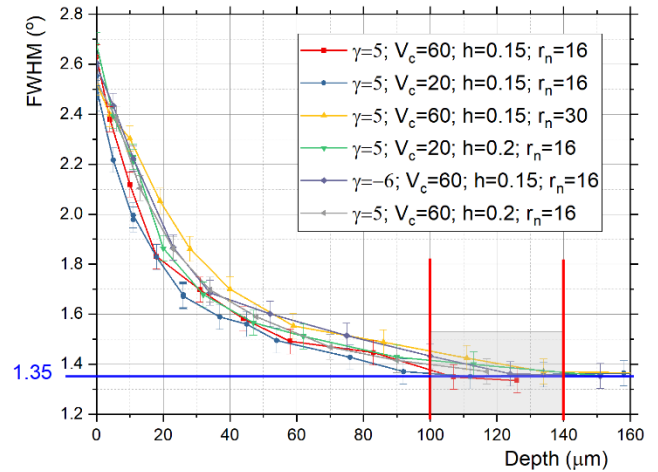


Fig. 6. Evolution of the FWHM of the X-ray diffraction peak in function of the depth beneath the machined surface.

For the six cutting conditions shown in Fig. 5a, the values of thicknesses are varying from 100 μm to 140 μm , which is highlighted by a grey region in Fig. 6. This thickness range agrees with the DIC measurements.

Equivalent plastic strain was also determined by numerical simulation. Fig. 7 shows the evolution of the equivalent plastic strain in function of the depth beneath the machined surface for several cutting conditions. The maximum strains near the surface are in the range of strain observed experimentally using DIC (Fig. 4). The depth at which the equivalent plastic strain reaches zero corresponds to the thickness of the plastically deformed layer induced by machining.

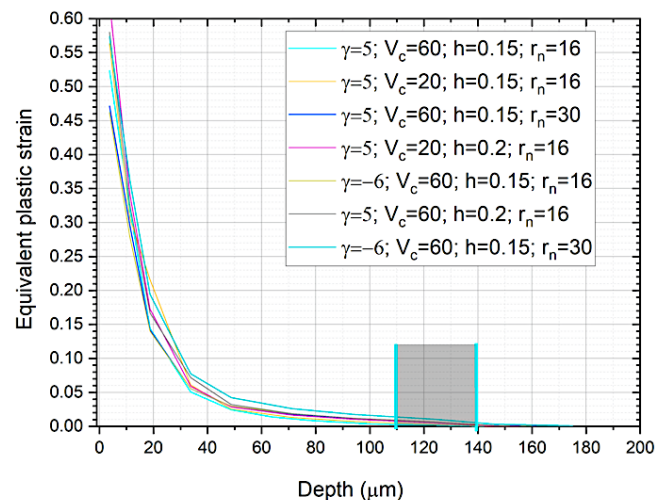


Fig. 7. Simulated profiles of residual equivalent plastic strain for different cutting conditions.

As can be seen in Fig. 7, these thicknesses are in the same range of the thicknesses determined experimentally (Fig. 5 and Fig. 6). The observed differences between the measured and predicted residual plastic strain profiles as shown in Fig. 8. can be attributed to the excessive distortion of the elements of the machined surface during the cutting simulation. These distorted elements affect the values of the near surface elements displacement and thus the resulting residual plastic deformation. Another reason is the tool-workpiece contact model that may not reproduce the exact contact between the tool and the machined surface. Moreover, the friction coefficients in function of the contact conditions (pressure, sliding speed, temperature, ...) need to be determined by tribological tests, for the same tool/workpiece materials pairs used in the machining tests.

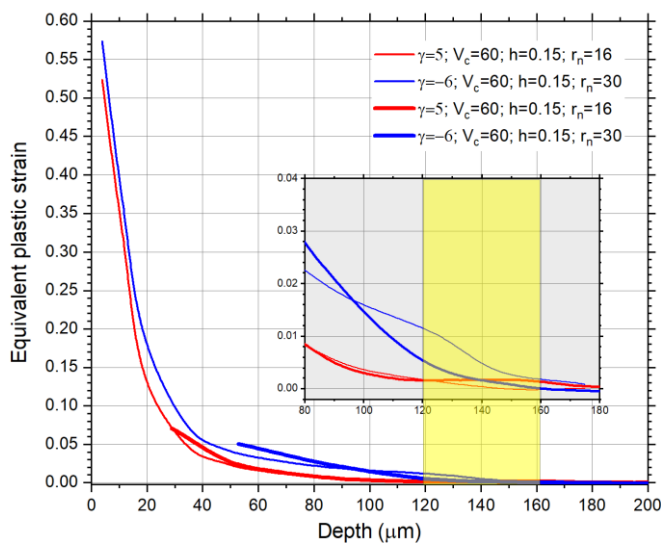


Fig. 8. Comparison between measured (thick lines) and simulated (thin lines) profiles of residual equivalent plastic strain for two cutting conditions.

5. Conclusion and outlook

The distribution of the plastic strain in the machined subsurface induced by orthogonal cutting of Ti-6Al-4V alloy were determined for several cutting conditions using a DIC. This technique could determine the strains for depth beneath the surface greater than 30 μm . To get the complete distribution of the plastic strain in the machined surface and subsurface, a 2D orthogonal cutting model of Ti-6Al-4V alloy was used. ABAQUS FEA software was used to FEM simulations using the Lagrangian formulation. Coupled thermo-mechanical analysis was performed to predict plastic strain distributions in the machined surface and subsurface.

The results have shown a strong gradient of plastic strain in the subsurface, with high plastic strains at the machined surface exceeding 50%. Experimental results obtained from selected cutting conditions were compared with those results obtained from 2D orthogonal cutting simulations. Both measured and predicted thicknesses of the deformed layers affected by machining are similar, as well as the distributions of plastic strain in the subsurface for depths greater than 30 μm . The determination of the plastic strain using DIC need to be improved to accurately determine the plastic strain immediately beneath the surface.

Acknowledgements

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